

CLEVER FIELD, OH, USA

WMO: 725224

Lat: 40.472N

Lon: 81.424W

Elev: 895

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.0	(c) 8.8	(d) -4.9	(e) 4.4	(f) 5.6	(g) -0.1	(h) 5.6	(i) 10.7	(j) 19.2	(k) 34.4	(l) 17.7	(m) 31.2	(n) 3.2	(o) 300	(p) 0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.7	73.0	87.5	72.1	84.8	70.9	76.1	85.0	74.6	83.0	73.2	81.2	6.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	127.3	80.8	72.1	122.8	79.3	70.6	116.6	77.5	40.0	85.3	38.6	83.0	37.3	81.2	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.8	(b) 14.7	(c) 12.4	(e) -4.2	(f) 92.0	(g) 6.5	(h) 3.1	(i) -8.9	(j) 94.2	(k) -12.8	(l) 96.0	(m) -16.4	(n) 97.7	(o) -21.2	(p) 100.0		
(4) 16.8	(5) 14.7	(6) 12.4	(7) -4.7	(8) 78.9	(9) 6.4	(10) 1.6	(11) -9.3	(12) 80.0	(13) -13.0	(14) 80.9	(15) -16.6	(16) 81.8	(17) -21.2	(18) 83.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	28.4	31.0	40.1	51.8	61.7	69.4	72.8	71.2	65.2	53.6	42.9	33.6		
	DBStd	18.00	12.34	11.32	10.84	9.17	8.09	6.06	5.16	4.96	7.03	8.89	9.28	10.08		
	HDD50	2492	677	541	347	86	8	0	0	0	1	63	252	517		
	HDD65	5558	1136	953	774	403	163	28	4	8	82	369	664	974		
	CDD50	3188	6	8	39	141	370	583	708	658	456	174	38	7		
	CDD65	782	0	0	1	8	60	162	247	201	87	16	0	0		
	CDH74	6917	0	1	14	134	632	1423	2143	1635	794	137	4	0		
	CDH80	2088	0	0	2	17	153	430	732	481	243	30	0	0		
(14) Wind	WSAvg	5.0	6.2	6.1	6.2	6.2	4.8	4.3	3.5	3.3	3.5	4.5	5.3	5.8		
(15) Precipitation	PrecAvg	40.8	3.0	2.0	2.9	3.5	4.4	4.3	4.4	3.9	3.5	2.8	3.1	3.0		
	PrecMax	59.4	6.8	4.7	7.0	5.2	8.1	8.5	8.0	7.9	7.4	4.6	5.4	8.0		
	PrecMin	31.2	1.3	0.8	1.2	1.8	2.0	1.1	2.0	1.1	1.5	0.7	1.1	1.3		
	PrecStd	7.3	1.4	0.9	1.3	1.0	1.7	2.0	1.6	1.7	1.7	1.1	1.1	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	66.8	75.2	82.2	88.1	90.6	93.1	91.0	90.5	84.5	72.1	64.2		
		MCWB	56.0	56.9	59.1	63.6	69.8	72.0	74.5	73.2	72.3	70.0	60.6	57.0		
	2%	DB	58.2	60.3	69.6	78.2	84.0	87.8	89.8	88.2	86.9	77.6	66.4	58.0		
		MCWB	53.1	52.5	56.6	61.1	67.7	71.7	74.2	73.0	70.8	65.0	56.8	52.2		
	5%	DB	52.4	54.3	63.7	73.6	81.1	84.8	87.1	85.2	82.2	73.1	62.9	54.2		
		MCWB	47.9	48.3	53.2	58.6	67.0	70.5	72.8	71.9	69.1	62.7	54.7	49.6		
	10%	DB	46.0	48.1	58.3	69.4	77.5	82.1	84.1	82.2	78.7	69.0	58.6	48.8		
		MCWB	41.6	42.5	50.1	56.3	64.8	69.6	71.5	70.5	66.8	60.7	51.7	44.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.8	58.5	61.1	67.2	73.0	76.0	78.4	77.6	75.6	71.5	63.3	58.7		
		MCDB	62.5	65.1	73.1	77.9	84.1	85.1	88.1	86.7	85.8	82.2	69.3	62.8		
	2%	WB	54.2	53.9	58.2	63.4	70.6	74.2	76.6	75.5	73.2	68.2	59.3	53.9		
		MCDB	58.2	58.6	66.7	73.3	80.4	83.5	85.8	83.7	81.9	74.6	64.7	57.1		
	5%	WB	48.5	48.4	55.2	61.0	68.6	72.8	74.9	73.9	71.3	64.8	55.7	50.0		
		MCDB	51.6	53.9	61.9	70.2	77.8	81.9	83.1	81.3	78.5	70.0	61.2	53.8		
	10%	WB	41.7	42.4	51.0	58.6	66.6	71.2	73.3	72.3	69.4	61.8	52.5	45.2		
		MCDB	46.1	48.0	57.3	67.1	75.0	79.5	80.8	79.3	75.6	67.9	58.1	49.0		
(35) Mean Daily Temperature Range	MDBR		15.2	16.8	19.3	22.6	21.9	20.6	20.0	20.4	21.8	20.5	18.2	14.2		
	5% DB	MCDBR	20.4	24.2	27.6	29.3	26.4	23.8	22.5	23.2	25.8	26.2	24.9	20.7		
		MCWBR	16.8	18.2	16.9	16.0	12.8	11.0	9.4	10.2	11.8	14.5	16.8	17.3		
	5% WB	MCDBR	19.2	22.9	24.3	24.0	22.1	20.0	19.1	18.6	20.3	21.4	20.6	19.5		
		MCWBR	17.2	18.7	16.2	15.3	11.7	10.1	9.2	9.5	10.1	12.8	15.8	17.7		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.326	0.358	0.391	0.432	0.459	0.462	0.454	0.393	0.364	0.321	0.306		
	taud		2.441	2.364	2.345	2.280	2.190	2.152	2.163	2.174	2.320	2.397	2.492	2.473		
	Ebn at Noon		272	281	283	279	269	261	259	258	268	263	264	262		
	Edh at Noon		25	31	36	41	46	48	47	45	36	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		481	752	1059	1437	1702	1848	1838	1687	1369	918	611	413		
	RadStd		42	102	106	144	141	142	130	89	125	83	85	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+371	+205
(47) Regional (42 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+122

Nomenclature: See separate page